Type MS · Relate sperm structures to their roles in fertilisation

Answer 1 of 8 · Question 6 part a.ii · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
6	(a) (i)	A = mitochondria; B = nucleus; C = acrosome;		3
	(ii)	Any three from • releases enzymes; • digestion of outer part of ovum/jelly coat; • to allow nucleus of sperm to enter; • reference to fertilisation;	Ignore contains / produces enzymes Ignore cell membrane Ignore head	3
	(iii)	• length on diagram is 122 (mm); • = 122 000(µm); • 122 000 ; • 50 • = 2440;	Allow a value between 120 and 122	4
	(b)	• release energy/produce ATP; • by (aerobic) respiration; • allows swimming/movement of sperm (to ovum);		3

Total 13 marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 1 of 8 · Question 6 part a.ii · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 2 of 8 · Question 5 part a.i · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
5	(a)	(i)	• join with ovum; • during fertilisation;		2
		(ii)	any two from • ribosomes; • endoplasmic reticulum/ER; • chromosomes;		2
	(b)		• site of (aerobic) respiration; • energy released/ATP produced; • sperm motile/active/swim;		3
	(c)		3µm = 0.003 mm; distance between X and Y = 145 mm; 145; 0.003 =48333/48000;	ECF = 2	4

Total 11 marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 2 of 8 · Question 5 part a.i · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 3 of 8 · Question 5 part b · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question number					
5	(a)	(i)	• join with ovum; • during fertilisation;		2
		(ii)	any two from • ribosomes; • endoplasmic reticulum/ER; • chromosomes;		2
	(b)		• site of (aerobic) respiration; • energy released/ATP produced; • sperm motile/active/swim;		3
	(c)		3µm = 0.003 mm; distance between X and Y = 145 mm; 145; 0.003 =48333/48000;	ECF = 2	4

Total 11 marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 3 of 8 · Question 5 part b · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 4 of 8 · Question 2 part a.i · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
2	(a)	(i)	movement/swimming;		1
		(ii)	C; A, B and D do not release energy		1
		(iii)	three from: • produces a haploid cell / reduces the number of chromosomes by half/to 23; • so at fertilisation the diploid/full number of/46 chromosomes is restored; • prevents doubling of chromosomes at fertilization; • introduces (genetic) variation;		3
	(b)	(i)	sperm egg X / Y X / XX / XY X / XX / XY ;	One for egg One for sperm One for the offspring	3
		(ii)	Two from: • acrosome; • contains enzymes; • digests/breaks down the egg cell membrane;		2

Total 10 marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 4 of 8 · Question 2 part a.i · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 5 of 8 · Question 2 part b.ii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
2	(a)	(i)	movement/swimming;		1
		(ii)	C; A, B and D do not release energy		1
		(iii)	three from: • produces a haploid cell / reduces the number of chromosomes by half/to 23; • so at fertilisation the diploid/full number of/46 chromosomes is restored; • prevents doubling of chromosomes at fertilization; • introduces (genetic) variation;		3
	(b)	(i)	sperm egg X / Y X / XX / XY X / XX / XY ;	One for egg One for sperm One for the offspring	3
		(ii)	Two from: • acrosome; • contains enzymes; • digests/breaks down the egg cell membrane;		2

Total 10 marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 5 of 8 · Question 2 part b.ii · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 6 of 8 · Question 3 part b.iii · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

3 (a)	• contains particular features / characteristics/ adaptations; • to perform a specific function;		2
(b) (i)	• 84 ÷ 0.006; • 14000x	Allow correct final answer from a calculation that includes 84 and 0.006, max 1 mark	2
(ii)	half the amount/haploid/23 chromosomes;		1
(iii)	• to provide energy; • for swimming/movement/travelling;		2
(iv)	• 180 ÷ 0.070; • 2571; • (÷ 60) 42.85;	Allow any figure that rounds to 43 Allow ecf for third mark - an incorrect value for a calculation carried out for mps1 and 2, divided by 60 **and** a correct answer to that calculation	3 exp
(c)	Any two from: • greater magnification; • higher resolution/can see objects more clearly/in more detail; • can see a greater number of structures;		2 exp

Total Question 3 = 12 marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 7 of 8 · Question 6 part a.i · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
6	(a) (i)	A = mitochondria; B = nucleus; C = acrosome;		3
	(ii)	Any three from • releases enzymes; • digestion of outer part of ovum/jelly coat; • to allow nucleus of sperm to enter; • reference to fertilisation;	Ignore contains / produces enzymes Ignore cell membrane Ignore head	3
	(iii)	• length on diagram is 122 (mm); • = 122 000(µm); • 122 000 ; • 50 • = 2440;	Allow a value between 120 and 122	4
	(b)	• release energy/produce ATP; • by (aerobic) respiration; • allows swimming/movement of sperm (to ovum);		3

Total 13 marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 7 of 8 · Question 6 part a.i · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 8 of 8 · Question 6 part b · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Question Number				
6	(a) (i)	A = mitochondria; B = nucleus; C = acrosome;		3
	(ii)	Any three from • releases enzymes; • digestion of outer part of ovum/jelly coat; • to allow nucleus of sperm to enter; • reference to fertilisation;	Ignore contains / produces enzymes Ignore cell membrane Ignore head	3
	(iii)	• length on diagram is 122 (mm); • = 122 000(µm); • 122 000 ; • 50 • = 2440;	Allow a value between 120 and 122	4
	(b)	• release energy/produce ATP; • by (aerobic) respiration; • allows swimming/movement of sperm (to ovum);		3

Total 13 marks

Type MS · Relate sperm structures to their roles in fertilisation

Answer 8 of 8 · Question 6 part b · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks